

1. Read the CSV File

- Load the dataset into a pandas DataFrame.
 - The actual column headers are on the **3rd row**, not the first.
 - Display the first 5 rows.
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2. Rename Columns

- Rename columns to clean format:
 - Remove spaces
 - Convert to lowercase
 - Example:
`Customer Name → customer_name`
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3. Drop Unnecessary Rows

- Remove the extra rows above the header.
 - Reset the index after cleaning.
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4. Drop Columns (if needed)

- Drop any column you think is not useful (justify your choice).
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5. Count Missing Values

- Count the number of null values in each column.
 - Display the result clearly.
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6. Handle Missing Values

a. Drop Rows Based on a Column

- Drop rows where `sales` is missing.

b. Fill Missing Values

- Fill missing values in:
 - `profit` using `mean`
 - `discount` using `median`
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7. Handle Missing Categorical Data

- Replace missing `customer_name` values with "Unknown"
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8. Detect and Remove Duplicates

- Identify duplicate rows based on:
 - `Orderid` (use `df.duplicated` to check if the value is duplicated or not)
- Remove duplicates and show how many were removed.

9. Filtering and creating new columns

- Display all orders where `unit_price` is greater than 20,000.
- display `customer_name`, `category`, and `status` for orders where `unit_price > 10,000` and `status == "Completed"`.
- Create a new column called `total_amount`:

`total_amount = quantity × unit_price`

- Create a column called `customer_type` based on quantity:
 - `Bulk Buyer` if `quantity >= 3`
 - `Regular Buyer` otherwise
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9. Final Clean Dataset

- Display:
 - Shape of dataset before and after cleaning
 - First 10 rows of cleaned dataset